



ICPC2T 2027



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SOCIETY

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2027 IEEE International Conference on Power, Control and Computing Technologies (ICPC2T 2027)

NIT Raipur, Chhattisgarh, India | 3–5 March 2027

National Institute of Technology Raipur — An Institute of National Importance

SPECIAL SESSION (SS4) ON **Intelligent Energy Optimization for Power, Control and** **Computing Systems**

LIST OF TOPICS

- Optimal power flow, unit commitment, economic dispatch, and congestion management
- Renewable-energy forecasting, scheduling, curtailment reduction, and grid integration
- Multi-objective, stochastic, robust, chance-constrained, and risk-aware optimization
- Model predictive control, adaptive control, reinforcement learning, and intelligent control
- Battery and hybrid energy-storage scheduling, sizing, control, and lifecycle-aware operation
- Power-electronic converter optimization, FACTS coordination, and power-quality enhancement
- Digital twins, IoT, edge/cloud computing, and real-time energy management systems
- Cyber-secure, resilient, and privacy-preserving optimization of energy systems
- Network reconfiguration, loss minimization, voltage control, and service restoration
- AI-, machine-learning-, and data-driven energy optimization
- Smart-grid, microgrid, and virtual power plant energy management
- Distributed, decentralized, hierarchical, and cooperative optimization
- Demand response, demand-side management, and flexible-load scheduling
- Electric-vehicle charging, fleet coordination, and vehicle-to-grid optimization
- Energy-efficient operation of buildings, campuses, industries, and data centres
- Metaheuristic, evolutionary, swarm, and hybrid computational optimization
- Energy markets, transactive energy, peer-to-peer trading, and price-responsive operation
- Real-time simulation, hardware-in-the-loop testing, benchmarks, and field applications

ORGANIZING CHAIRS



Dr. Sheila Mahapatra

Alliance University, Bangalore



Dr. Hemani Kaushal

University of North Florida, USA

SUBMISSION DETAILS

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After login, select Special Session (SS4): “Intelligent Energy Optimization for Power, Control and Computing Systems”

CONTACT INFORMATION

sheila.mahapatra@alliance.edu.in

hemani.kaushal@unf.edu